

SOC Incident Detection & Log Analysis System

Project Report

Synopsis

This project automates the collection, analysis, and monitoring of security logs to detect suspicious activities such as failed logins, brute-force attacks, unauthorized access attempts, and unusual network behavior. The system generates alerts and reports to support Security Operations Center (SOC) activities.

Objectives

- Collect logs from multiple sources.
- Detect suspicious and malicious activities.
- Generate security alerts.
- Provide incident investigation reports.
- Improve security monitoring and threat detection.

Software Requirements

Operating System: Windows 11 / Linux

Programming Language: Python

IDE: Visual Studio Code

Database: SQLite

Libraries: Pandas, Regex, Matplotlib

Features

- Log Collection and Processing
- Failed Login Detection
- Brute Force Attack Detection
- Suspicious IP Monitoring
- Incident Reporting
- Security Alert Generation

Modules

User Input Module, Log Collection Module, Detection Module, Alert Generation Module, and Reporting Module.

Sample Detections

- Multiple Failed Logins – Medium Severity
- Brute Force Attack – High Severity
- Unauthorized Access – High Severity
- Suspicious IP Activity – Medium Severity
- Privilege Escalation – Critical Severity

Conclusion

The SOC Incident Detection & Log Analysis System helps security analysts identify threats quickly by automating log analysis, alert generation, and incident reporting.